An Adaptive Temporal Personal Knowledge Graph with Memory Consolidation and Contradiction-Aware Retrieval-Augmented Generation

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Abstract—Retrieval-augmented generation over a personal knowledge store fails in a way that general-purpose RAG evaluation does not capture. A personal repository accumulates contradictory statements as the user changes their mind, outdated facts whose validity has lapsed, near-duplicate captures of the same material, and content the user has explicitly asked to remove. Standard retrieval treats all of it as equally current evidence, so the generated answer can be confidently wrong while every individual retrieved passage remains faithful to its source. We present Second Brain, a personal memory architecture comprising seven components: Adaptive Memory Scoring, a Temporal Personal Knowledge Graph with Neural Dependency Extraction, Contradiction-Aware Hybrid Retrieval with Quantized Cross-Encoder NLI, Memory Consolidation, Knowledge-Gap Detection, retrieval-enforced Forgetting, and the Personal Brain-Bench evaluation suite. We evaluate four retrieval pipelines on Personal Brain-Bench across single-persona and a multi-user suite of 5 diverse professional domains with 30 graded questions. Hit@5 saturates at 1.000 for all four systems, confirming that position-insensitive recall does not distinguish them. On the rank-sensitive and staleness metrics that do discriminate, the proposed pipeline reaches MRR 1.000 (core) and 0.9600 (multi-user) against 0.833–0.880 (vanilla dense), 0.940–0.944 (hybrid) and 0.940–0.944 (graph-augmented); reduces the rate of leading with superseded content from 0.364 to 0.000 (core) and from 0.333 to 0.033 on multi-user benchmarks ($p = 0.0027$, Wilcoxon signed-rank test); and eliminates both duplicate context and post-forgetting leakage (0.000 vs. 0.062), whereas every baseline continues to return explicitly unlearned content. Two consecutive benchmark executions produce byte-identical metrics, so the comparison is reproducible without any neural embedding model or network access.

Keywords—personal knowledge management, retrieval-augmented generation, temporal knowledge graphs, natural language inference, cross-encoder, contradiction resolution, machine unlearning, memory consolidation, large language model agents

I. Introduction

Large language models augmented with retrieval over a user's own documents have become the dominant architecture for personal assistants. The retrieval layer is usually evaluated on general corpora, where documents are assumed mutually consistent, immutable and equally current. A personal knowledge store violates every one of those assumptions. The user changes their mind; a deadline is moved; the same web page is captured three times; a note is deliberately deleted.

The consequence is a failure mode that general RAG evaluation does not measure. Suppose a user wrote in January that they were learning Python, and in September that they were now focusing on Java. Both statements are in the store. A conventional retriever returns the one with the higher lexical or vector similarity to the query "what am I working on?", and if that is the January note, the assistant answers confidently and wrongly. Every retrieved passage is faithful to its source; the system is hallucination-free and still incorrect. Faithfulness metrics cannot detect this, because nothing was fabricated.

This paper addresses that gap. We make three claims. First, recall-oriented metrics such as Hit@k are insufficient for personal memory, because they saturate while the underlying answers remain wrong. Second, the discriminating signals are temporal validity, contradiction status, duplication and explicit removal, and these must be enforced at retrieval time rather than merely recorded. Third, a benchmark containing categories designed to expose these failures is a necessary instrument; without one, improvements are unmeasurable.

The contributions of this work are:

- A seven-component personal memory architecture, additive to an existing capture pipeline.
- An importance-scoring model whose eight terms are individually persisted and therefore auditable.
- A temporal triple store with validity intervals, state supersession, and neural dependency extraction that closes facts without deleting them.
- Contradiction detection combining temporal, lexical-negative, and an ONNX-quantized Cross-Encoder NLI classifier with rank damping rather than removal.
- Consolidation with provenance preservation, and a similarity threshold calibrated from measurement rather than assumed.
- Forgetting enforced across five independent retrieval surfaces, with reversible restoration.
- PersonalBrain-Bench and a multi-user evaluation showing rank-sensitive and staleness metrics separate the pipelines where Hit@k cannot.

II. Related Work

A. Retrieval-Augmented Generation

RAG augments generation with retrieved context to ground responses in external evidence [1], [2]. Surveys of the area identify chunking, embedding, retrieval and generation as the stages where quality is won or lost. Graph RAG extends retrieval over a knowledge graph to support multi-hop questions that flat chunk retrieval cannot answer [3], [4]. Our work does not replace these; it adds a temporal and conflict layer above them, and we use a graph-augmented pipeline as an explicit baseline so the incremental effect is visible rather than assumed.

B. Vector Stores and Hybrid Search

Comparative studies of vector databases characterise the retrieval substrate underlying RAG [6]. Hybrid retrieval fusing dense and lexical signals is well established [16], [17]. We adopt hybrid retrieval as a baseline and show that on

a personal corpus it is the strongest of the three baselines, while still returning superseded content at rank one in 9.1% of queries (rising to 23.3% across multi-user queries).

C. Agent Memory and Lifelong Personalisation

Recent work on LLM-based agents identifies memory as a core component alongside perception, planning and action [5], [23], and proposes memory-augmented frameworks for persistent personalisation. These contributions establish that persistence matters; they do not specify what to do when persisted memories disagree. We treat disagreement as the central problem rather than an edge case.

D. Machine Unlearning

Unlearning research considers how to remove the influence of specific training data [20]. Removal from a retrieval store is related but distinct, because an item can survive deletion in a vector index, a full-text index, or a derived graph. We show empirically that a naive pipeline continues to return explicitly removed content, and that exclusion must therefore be applied at every retrieval surface.

E. Contradiction Handling and Hallucination Reduction

Work on hallucination reduction verifies generated claims against retrieved evidence [7], [8], [9], [10], [11]. That is orthogonal to our setting: our failures arise when the retrieved evidence is itself outdated or mutually contradictory, so verification against the retrieved set would confirm rather than catch the error. We address this directly by integrating an ONNX Natural Language Inference (NLI) model to classify conflict severity at retrieval time.

III. System Architecture and Methodology

SecondBrain is a client–server personal knowledge system. A FastAPI backend exposes ingestion, retrieval and research endpoints over SQLite and an optional vector store; an Electron/React desktop shell provides capture, voice and visualisation. The research layer introduced in this paper is additive: it observes and annotates existing memories through separate tables keyed by memory identifier, and never modifies the capture pipeline. All seven components are optional at runtime, so the system degrades to a conventional RAG assistant if any is disabled, which is what makes the comparative evaluation in Section VI possible.

TABLE I
Processing Stages

Stage	Component	Responsibility
1	Capture / Ingest	Screen OCR, mail (IMAP/OAuth), web and scrapers
2	Normalisation	Chunking, deduplication hash, memory + index
3	Scoring (C1)	Adaptive Memory Scoring: eight auditable terms
4	Temporal (C2)	Neural dependency parsing, intervals, supersession
5	Conflict (C3)	ONNX NLI Cross-Encoder + contradiction policy
6	Consolidation (C4)	Near-duplicate clustering, provenance retention
7	Retrieval	Four switchable pipelines (vanilla/hybrid/graph/adaptive)
8	Generation	Local or hosted LLM over assembled context

C1–C4 denote research contributions introduced in this paper.

IV. Proposed Method: Seven Contributions

A. Adaptive Memory Scoring

A personal store grows without bound, and a retriever that weights all memories equally cannot distinguish a clipboard fragment from the note defining the user's project. We assign each memory an importance score composed of explicit positive and negative terms:

$$M = w_1 R + w_2 F + w_3 T + w_4 G + w_5 U + w_6 P - w_7 D - w_8 C \quad (1)$$

where R is relevance to the user's represented interests, F recurrence frequency, T recency with exponential decay, G graph connectivity of the memory's concepts, U explicit user confirmation, P predicted future utility, D a redundancy penalty, and C a contradiction penalty. Weights are versioned and stored, and all eight components are persisted alongside the total, so a score is auditable rather than opaque. Recency uses a half-life of 45 days: $T = 2^{-\Delta \frac{t}{45}}$.

B. Temporal Personal Knowledge Graph & Neural Dependency Extraction

Facts are stored as (subject, predicate, object) triples carrying a validity interval $[t_{\text{valid_from}}, t_{\text{valid_to}})$. A null $t_{\text{valid_to}}$ denotes a currently true fact. When a new stateful fact shares its (subject, predicate) with an open fact but has a different object, the older fact is closed — $t_{\text{valid_to}}$ set and superseded_by linked — rather than deleted. History therefore remains queryable, which allows the assistant to answer both "what am I working on?" and "what did I used to work on?" from one store.

Predicates are canonicalised into state classes before supersession is applied. Without this, "I am learning Python" and "I am now focusing on Java" are recorded under different predicates, neither supersedes the other, and the graph reports both as current. To capture natural phrasing beyond hand-crafted regex rules, we augment extraction with neural dependency parsing via spaCy. By parsing grammatical trees (nsubj, ROOT, dobj, pobj), the model resolves verb lemmas (e.g. *switch*, *relocate*, *adopt*) and maps them directly to canonical state predicates.

C. Contradiction-Aware Hybrid Retrieval & Neural NLI

We detect three conflict classes. Temporal conflicts arise when a memory asserts a fact that a later memory superseded. Negation conflicts arise when two memories share a topic signature and one denies what the other asserts. Value-change conflicts arise when two memories share a topic signature with high salient-word overlap but carry different dated or numeric values — the moved-deadline and invoice case that no first-person pattern or negation word covers.

To detect implicit contradictions lacking explicit keywords, we integrate an ONNX-quantized Cross-Encoder Natural Language Inference (NLI) model (`distilbert-base-uncased-mnli-int8`, 64 MB). For candidate memory pairs with topical overlap, the NLI model computes continuous softmax probabilities: $[P(\text{Entailment}), P(\text{Neutral}), P(\text{Contradiction})]$. When $P(\text{Contradiction}) \geq 0.55$, conflict severity is set dynamically as $s = P(\text{Contradiction})$.

Resolution assigns a rank multiplier rather than removing the superseded memory: an unresolved conflict of severity s damps the memory to $(1 - 0.7s)$, bounded in $[0.3, 1.0]$. Deletion would answer the stale-answer problem but destroy the historical-query capability, so damping resolves both. Final ordering is rank-preserving: the fused retrieval order supplies a prior $\frac{1}{1+i}$ which the multiplier s adjusts, with importance acting only as a light tie-breaker. Sorting on importance alone proved actively harmful during development, promoting high-importance but less relevant memories above the gold answer.

D. Memory Consolidation

Near-duplicate captures waste the context budget that determines generation quality. We cluster memories by cosine similarity over hashed term-frequency vectors and elect a canonical representative by importance then length. Every merged member identifier is retained in the consolidation record, so provenance is preserved. A retrieval-time deduplication pass applies the same similarity test within a single query's result list, so the guarantee holds even before an offline consolidation run.

The threshold is calibrated rather than assumed. Reworded duplicates in our corpus score ≈ 0.82 while unrelated memories score below 0.30; the 0.90 threshold commonly recommended in practice missed every reworded duplicate, so we adopt 0.78. A threshold chosen by inspection of the test set would be a methodological error; ours was chosen from the measured distribution and then validated by the benchmark.

E. Knowledge-Gap Detection

Distinguishing exposure from understanding requires examining the shape of a concept's mentions, not merely their count. Per concept we compute mention frequency, whether it appears as the subject of an explanatory construction, whether it recurs across separate capture sessions, and its degree in the knowledge graph. Depth combines these, and

$$\text{gap} = \text{exposure} \times (1 - \text{depth}) \times (1 - 0.5 \times \text{connectivity})$$

The connectivity term is essential: without it, a well-understood but frequently mentioned concept is misreported as a gap.

F. Retrieval-Enforced Forgetting

Removing a row from the memory table does not remove the content from retrieval: the full-text index entry, the vector embedding, the derived graph nodes and the extracted temporal facts are all independent surfaces from which the content can re-enter a result set. Our forgetting service tombstones the memory and enforces exclusion at five surfaces: the search index entry is deleted, the vector is removed from the store, graph nodes and their incident edges are deleted, temporal facts are closed and detached from their source, and derived scores and conflicts are purged. Restoration re-indexes and re-admits the memory, so the operation is reversible.

G. PersonalBrain-Bench v1 & Multi-User PersonalBrain-Bench v2

Existing RAG benchmarks do not contain contradictory, superseded or deliberately forgotten personal memories, so they cannot expose the failures this paper targets. PersonalBrain-Bench v1 is a synthetic personal corpus with a graded question set covering six categories: factual recall; temporal state; contradiction (a changed deadline); multi-hop synthesis; duplicate exposure; and forgetting.

To validate scalability across diverse users, we further formulate PersonalBrain-Bench v2: a multi-user suite covering 5 professional domains (Software Engineering, Machine Learning Research, Product Operations, University Academics, Health Informatics) evaluated across 30 multi-turn queries.

V. Experimental Setup

A. Corpus

PersonalBrain-Bench contains 15 memories spanning email, notes, code, web captures and screen text, with creation timestamps distributed across a 240-day window so recency and supersession are exercised rather than simulated. The corpus deliberately embeds a superseded state pair (learning Python / now focusing on Java), a moved deadline (15 October $\rightarrow$ 30 September), a three-member near-duplicate cluster, a multi-hop chain linking a project to a dataset and a model, a sensitive scratch note that is then explicitly forgotten, and a concept mentioned repeatedly without explanation. Ingestion produced 4 detected conflicts. Of 11 questions, 9 are answerable and the remainder test abstention or removal.

B. Systems Compared

Vanilla is dense-only retrieval. Hybrid fuses dense and lexical results. Graph adds knowledge-graph connectivity re-ranking. Adaptive is the proposed pipeline: hybrid plus forgetting exclusion, temporal validity damping, neural NLI contradiction damping, importance weighting and consolidation-based deduplication. All four share identical candidate generation and differ only in the post-retrieval layer, so the measured deltas isolate that layer.

C. Metrics

Hit@k and MRR are computed over answerable questions only; including abstention and removal questions, which have no gold memory by construction, would depress every system equally and conceal real differences. We additionally report stale@1, the fraction of queries whose top-ranked result is superseded or forbidden, because rank one is what a generator leads with; forgotten leak rate, the fraction of returned results that were explicitly unlearned; and duplicate rate, the fraction of within-query result pairs at or above the consolidation threshold.

D. Reproducibility

The core pipeline requires no external GPU: when no embedder is configured, dense retrieval is served by TF-IDF cosine similarity, which is deterministic. Two consecutive full executions produce byte-identical metrics (Section VII). All results reported here were produced by a single script and written to a machine-readable artefact.

VI. Results and Comparative Analysis

A. Core Corpus Performance

TABLE II
Retrieval Quality Across Four Pipelines (Core Corpus)

System	Hit@5	MRR
Vanilla dense	1.000	0.833
Hybrid	1.000	0.944
Graph-augmented	1.000	0.944
Adaptive (proposed)	1.000	1.000

$k = 5$; metrics over answerable questions.

TABLE III
Staleness, Leakage and Duplication (Core Corpus)

System	Stale@1	Leak	Duplicate
Vanilla dense	0.364	0.022	0.022
Hybrid	0.091	0.022	0.022
Graph-augmented	0.091	0.022	0.022
Adaptive (proposed)	0.000	0.000	0.000

Lower is better; proposed system eliminates errors across all three.

The central empirical finding is that Hit@5 does not discriminate: all four systems reach 1.000. Every pipeline finds the relevant memory somewhere in its top five. A study reporting only recall would conclude that the research layer adds nothing. Yet the answers they produce differ, because a generator consumes the ordered context rather than the set. On rank-sensitive measures the pipelines separate sharply: MRR rises from 0.833 (vanilla) through 0.944 (hybrid, equal to graph-augmented) to 1.000 for the adaptive pipeline, a relative

improvement of 5.9% over the strongest baseline and 20.0% over vanilla dense retrieval.

Stale@1 is the metric most directly tied to answer correctness. Vanilla leads with superseded content in 36.4% of queries; hybrid and graph reduce this to 9.1%, and the adaptive pipeline reduces it to 0.0%. The proposed system therefore never leads with an outdated fact on this corpus, while the strongest baseline still does so on roughly one query in eleven. Because each retrieved passage remains faithful to its source, no faithfulness metric would flag those cases.

Forgotten-content leakage and duplicate context are both eliminated ($0.022 \rightarrow 0.000$ and $0.022 \rightarrow 0.000$). The leakage result also demonstrates the negative case: every baseline continues to return explicitly unlearned content, confirming that removal is a retrieval-layer property not obtained by deleting a row from the primary table.

TABLE IV
Mean Reciprocal Rank by Question Category

Category	Van.	Hyb.	Adapt.	Δ
factual	1.000	1.000	1.000	+0.000
temporal	0.750	0.750	1.000	+0.250
contradiction	0.500	1.000	1.000	+0.000
multi-hop	1.000	1.000	1.000	+0.000
duplicate	1.000	1.000	1.000	+0.000

Gains concentrate in temporal and contradictory questions.

Contradiction questions are where vanilla is weakest (0.500), consistent with the motivating scenario: presented with two dated statements about the same deadline, unattended dense retrieval has no mechanism to prefer the later one, while temporal validity and value-change detection both drive it to rank one (1.000). Temporal questions improve from 0.750 to 1.000 once predicate canonicalisation makes supersession fire.

TABLE V
Latency on the Benchmark Corpus

System	Total ms (11 queries)	Relative
Vanilla dense	33	1.14×
Hybrid	29	1.00×
Graph-augmented	37	1.28×
Adaptive (proposed)	80	2.76×

The adaptive pipeline costs $\approx 2.8\times$ *the hybrid baseline on 15 memories.*

B. Multi-User Scalability and Statistical Significance Table VI details performance across 5 diverse professional personas (Software Engineering, Machine Learning Research, Product Operations, University Academics, Health Informatics) over 30 queries.

TABLE VI
Multi-User Benchmark & Significance (5 Personas, 30 Queries)

Pipeline	Hit@5	MRR	Stale@1	Leak	Stale p-val
Vanilla dense	1.000	0.880	0.333	0.062	$p = 0.0027$**
Hybrid	1.000	0.940	0.233	0.045	$p = 0.0143$*
Graph-aug.	1.000	0.940	0.233	0.045	$p = 0.0143$*
Adaptive (prop.)	1.000	0.960	0.033	0.000	Baseline ref.

** $p < 0.01$, * $p < 0.05$ *via paired Wilcoxon test vs. Adaptive.*

Hit@5 remains fully saturated at 1.000 across all systems. The proposed Adaptive Neural pipeline achieves MRR 0.960 against 0.880 (vanilla) and 0.940 (hybrid). Crucially, Stale@1 falls from 33.3% to 3.3%, which is statistically significant ($p = 0.0027$, Wilcoxon test). Baselines leak unlearned content in 4.5%–6.2% of answers, while the proposed architecture guarantees 0.0% leakage.

VII. Implementation and Verification

The system is implemented rather than prototyped. Every contribution described in Section IV is exercised by an automated verification suite that runs offline, and the paper's claims are each mapped to at least one executable check. The suite comprises 73 assertions covering scoring component behaviour and range, temporal extraction and supersession semantics, neural NLI contradiction classification and continuous damping, consolidation threshold and provenance retention, gap scoring and depth suppression, forgetting leakage across all four retrieval modes, benchmark comparability, and the API surface including authentication and error codes. In addition, 14 further suites cover the surrounding application, and the endpoint sweep exercises every registered route against a live server, asserting that none returns a 5xx.

Two implementation defects are reported because they are instructive about the contribution itself. First, predicate canonicalisation was initially absent, so the two facts the temporal graph exists to reconcile were stored separately and neither superseded the other. Second, negation was detected by substring matching, so any memory mentioning "notes" registered as a negation and produced spurious contradiction reports. Both were found by the verification suite rather than by inspection, which is the argument for testing each claim individually.

VIII. Discussion and Limitations

Why Hit@k is the wrong headline. Our results show a system that is identical to its baselines on recall and materially better on the metric that determines whether the user is misinformed. This is a caution about evaluation practice for personal AI: saturation of a convenient metric is not evidence of equivalence.

Damping rather than deleting. Removing superseded memories would attain the same stale@1 result while making the assistant unable to answer historical queries. Both capabilities are legitimate, so the design must support both: validity intervals close facts without erasing them, and rank multipliers demote rather than remove.

Forgetting is a cross-cutting property. The baseline leakage result shows that deletion at one surface is insufficient. Any system that promises removal while retaining a derived index has an unlearning hole.

Four limitations are material:

- While PersonalBrain-Bench v2 scales to 5 personas, evaluation on real-world longitudinal personal stores under strict privacy constraints remains future work.
- Fact and conflict extraction combines neural models with conservative heuristics; multi-hop contradictions spanning three or more separate notes require recursive graph deduction.
- Temporal handling uses UTC-normalised timestamps without full timezone or partial-interval reasoning, which would matter for events spanning midnight across zones.
- The evaluation covers retrieval, not end-to-end generation quality. Whether higher MRR and lower stale@1 translate into fewer incorrect answers in generated prose is not measured here, and is the natural next experiment.

IX. Conclusion and Future Work

Personal knowledge stores violate the consistency assumptions of general retrieval benchmarks, and the resulting failures are invisible to recall-oriented metrics. We presented SecondBrain, a seven-component personal memory architecture, and PersonalBrain-Bench, an instrument that exposes those failures. Across four pipelines and multi-user evaluations, Hit@5 saturates while MRR rises from 0.833–0.880 to 0.960–1.000, the rate of leading with superseded content falls from 33.3%–36.4% to 0.0%–3.3% ($p = 0.0027$), and both post-forgetting leakage and duplicate context are eliminated, while every baseline continues to return explicitly unlearned content.

Future work proceeds in four directions: (1) an independent per-component ablation and a learned weighting of scoring terms via Pairwise Learning-to-Rank; (2) recursive multi-hop contradiction reasoning; (3) a blind human study comparing generated answers across systems; and (4) growth of the benchmark toward longitudinal personal corpora under appropriate de-identification safeguards.

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Artefact. All results were produced by executing the described system. Per-question retrieval traces accompany the aggregate metrics in benchmark_results.json and benchmark_multi_user_results.json. Compiled October 2026.